Worksheet 30 · Sketching Graphs 2 IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Draw graphs of cubic and reciprocal functions • Read values and estimate gradients from a graph • Use graphs to find approximate solutions of equations		
Grade: / 23	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Sketching Graphs 2*

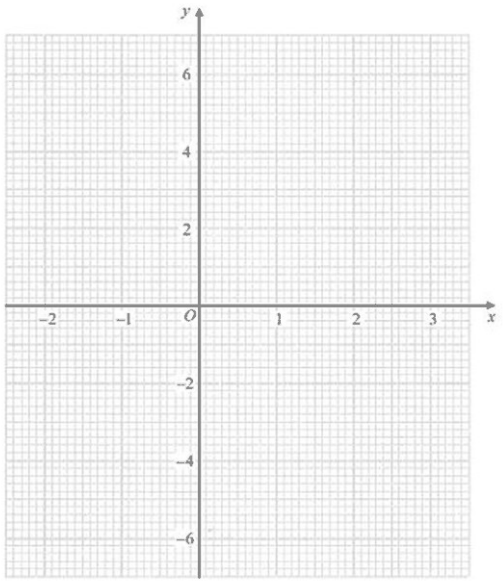
Question 1 (8 marks)

(a) Complete the table of values for $y = x^3 - 2x^2 - 3x + 4$

x	−2	−1	−0.5	0	1	1.5	2	3
y			4.875	4		−1.625		

(2)

) On the grid, draw the graph of $y = x^3 - 2x^2 - 3x + 4$ for values of x from −2 to 3



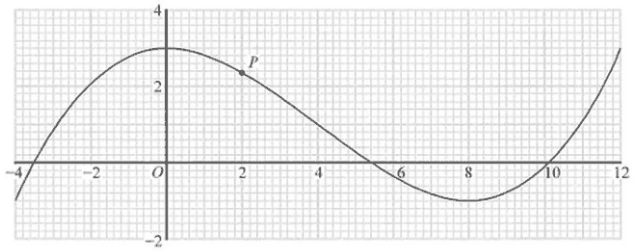
(2)

c) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^3 - 2x^2 - x + 1 = 0$
Give your solutions correct to 1 decimal place.

(4)

Question 2 (7 marks)

The diagram shows the graph of $y = f(x)$ for $-4 \leqslant x \leqslant 12$



The point P on the curve has x coordinate 2

(a) (i) Use the graph to find an estimate for the gradient of the curve at P.

(3)

(ii) Hence find an equation of the tangent to the curve at P.
Give your answer in the form $y = mx + c$

(2)

The equation $f(x) = k$ has exactly two different solutions for $-4 \leqslant x \leqslant 12$

(b) Use the graph to find the two possible values of k.

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(2)

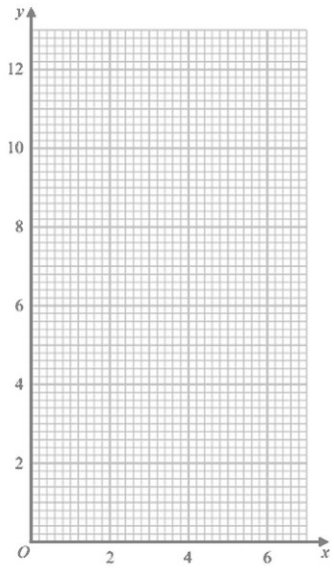
Question 3 (4 marks)

(a) Complete the table of values for $y = \frac{6}{x}$

x	0.5	1	2	3	4	5	6
y		6		2			1

(2)

(b) On the grid, draw the graph of $y = \frac{6}{x}$ for $0.5 \leqslant x \leqslant 6$



(2)

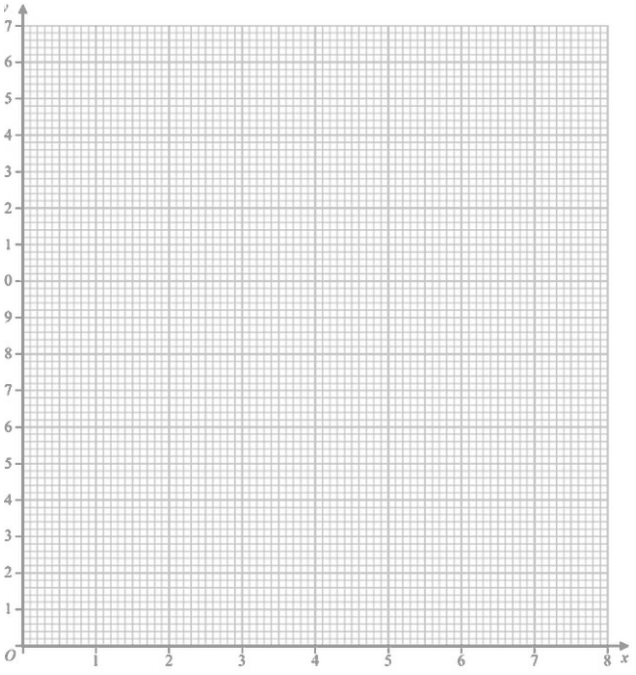
Question 4 (4 marks)

(a) Complete the table of values for $y = \frac{1}{x}(x^2 + 4)$

x	0.25	0.5	1	2	4	8
y	16.25					8.5

(2)

(b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leqslant x \leqslant 8$



(2)